

AETHYR VILLELA

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EDUCATION

The University of Texas at Austin, Austin, TX

May 2026

Bachelor of Science, Biomedical Engineering, Concentration in Instrumentation & Imaging

Certificate in Forensic Science

Relevant Coursework: Biomedical Engineering Signals and Systems Analysis, Micro and Nanotechnology for Sensors, Mechatronics

SKILLS

Technical: Python, MATLAB, C++, R, Fusion360, SolidWorks, MS Office

Certifications: 3D Printing, Laser Cutting, IBM Project Manager

Honors: Friends of Alec Engineering Scholarship, FHL Bank of Dallas Scholarship Recipient, 2022 Hispanic Scholarship Foundation Scholar, ULN Scholar, Ramshorn Program Scholar, Texas Challenge Scholarship

EXPERIENCE

SurfPlasma, *Business Development Intern*

Jun 2025 - Present

- Collaborate with R&D and commercialization leadership to align the product roadmap with market needs and regulatory pathways
- Support regulatory strategy by organizing documentation in compliance with ISO, FDA, OSHA, and EPA standards
- Conduct strategic outreach to identify and engage potential B2B partners, generating leads for early commercialization

Project Advance Austin (PAA), *Project Management Intern*

Aug 2024 - May 2025

- Collaborated with SpineHope and a team of 3 other UT students to design an educational curriculum, website, and interactive activities to simplify spinal conditions for children and promote body positivity
- Earned IBM Project Manager Certification, showcasing expertise in managing project timelines, tasks, and communication effectively

Crown Castle, *AI/ML May Intern*

May 2024

- Developed a computer vision AI/ML model using Python, Docker, Linux, and TensorFlow to detect brain tumors with a 71% validation rate
- Utilized a dataset of annotated MRI images and implemented a convolutional neural network (CNN) architecture

Worcester Polytechnic Institute (WPI), *Engineering Design Intern*

May 2023 - Jul 2023

- Collaborated with a team of 3 biomedical engineering students to design solutions for global health challenges
- Conducted market research, finding a needs gap and investigating
- Developed a Portable Water Purification System by utilizing SolidWorks

PROJECTS

2025 National Medical Device Make-A-Thon Competition

Feb 2025

- Placed 6th nationwide and won First Place prelim round with a low-intensity ultrasound device for osteoarthritis; delivered regulatory and market analysis

LEADERSHIP & COMMUNITY INVOLVEMENT

Biomedical Engineering Society (BMES), *Vice President of External Affairs*

May 2024 - Present

- Spearheaded a new program, BMESim, enabling BMES members to view, participate, and conduct outreach through Dell Children's medical simulation program
- Executed a Case Competition – increased team participation by 67% compared to last year
- Organized a Career Night panel, contributing to the professional development of over 150 BMES members

Ramshorn Scholars Program (RSP), *Manager*

May 2024 - Present

- Oversee a team of 45 students, providing guidance to 150 first-year engineers while applying project management and process improvements to enhance program outcomes
- Assist voter registration of Ramshorn Scholars through directed initiatives and collaborations
- Organize 6 socials throughout the academic year, increasing engagement and student retention